

# STATISTICAL BUSINESS ANALYSIS REPORT

## 1. Project Overview

This project analyzes sales and customer churn datasets using statistical methods. The report includes descriptive statistics, distribution analysis, correlation analysis, hypothesis testing, confidence intervals, and regression analysis.

## 2. Dataset Information

Sales Dataset Shape: (100, 7)

Customer Churn Dataset Shape: (500, 9)

## 3. Descriptive Statistics

Average Sales: 123650.48

Median Sales: 97955.50

Mode Sales: 6540.00

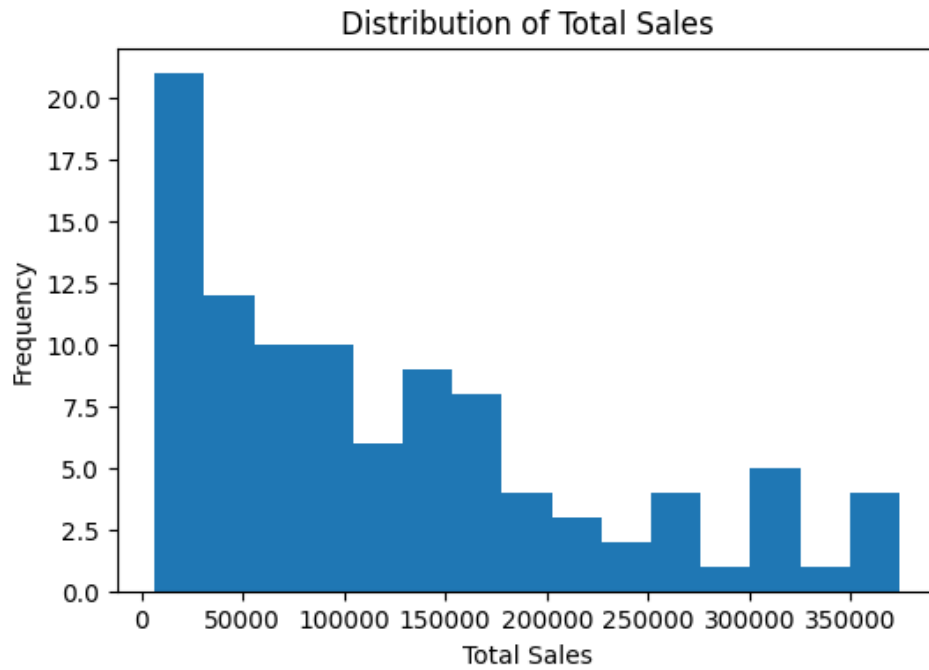
Standard Deviation: 100161.09

| Statistic | Quantity | Price    | Total_Sales |
|-----------|----------|----------|-------------|
| count     | 100.00   | 100.00   | 100.00      |
| mean      | 4.78     | 25808.51 | 123650.48   |
| std       | 2.59     | 13917.63 | 100161.09   |
| min       | 1.00     | 1308.00  | 6540.00     |
| 25%       | 2.75     | 14965.25 | 39517.50    |
| 50%       | 5.00     | 24192.00 | 97955.50    |
| 75%       | 7.00     | 38682.25 | 175792.50   |
| max       | 9.00     | 49930.00 | 373932.00   |

## 4. Data Distribution Analysis

Shapiro-Wilk Test P-value: 0.00000

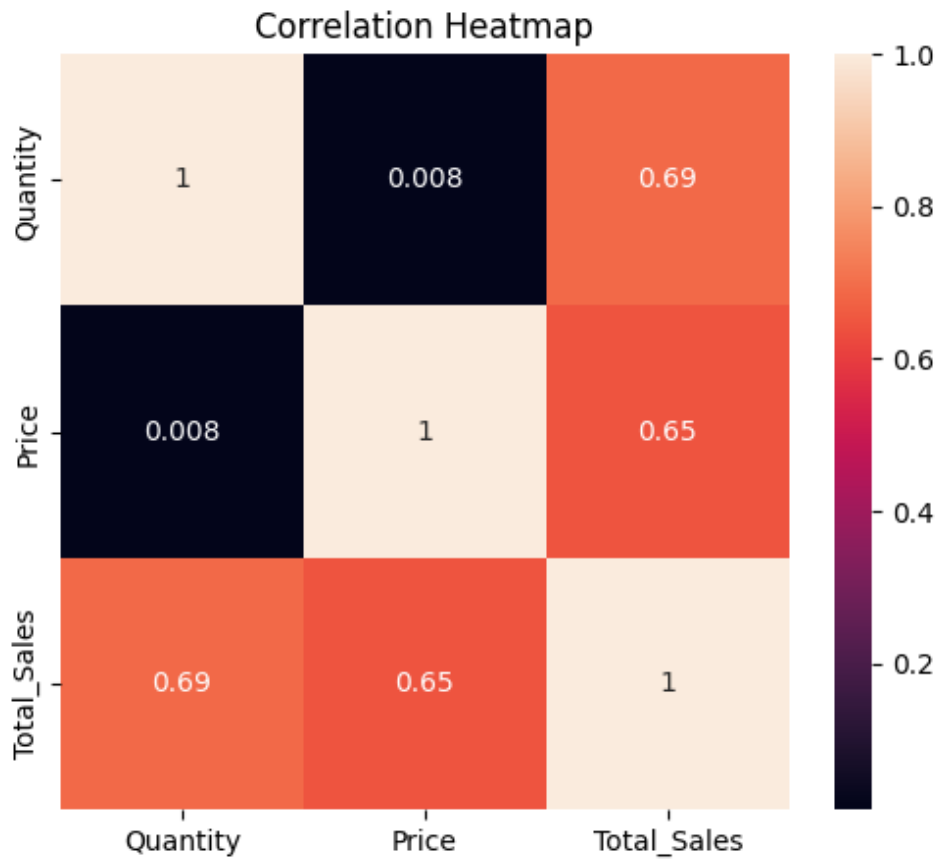
The sales data is not normally distributed.



## 5. Correlation Analysis

Pearson Correlation (Quantity vs Total Sales): 0.69

Moderate positive correlation exists.



## 6. Hypothesis Testing

### Test 1: Independent T-Test

P-value: 0.04962

Result: Significant difference exists between East and West regional sales.

### Test 2: ANOVA Test

P-value: 0.15329

Result: Product sales do not differ significantly.

### Test 3: One Sample T-Test

P-value: 0.00000

Result: Average monthly charges are significantly different from 100.

## 7. Confidence Interval

95% Confidence Interval for Average Sales: (103776.35, 143524.61)

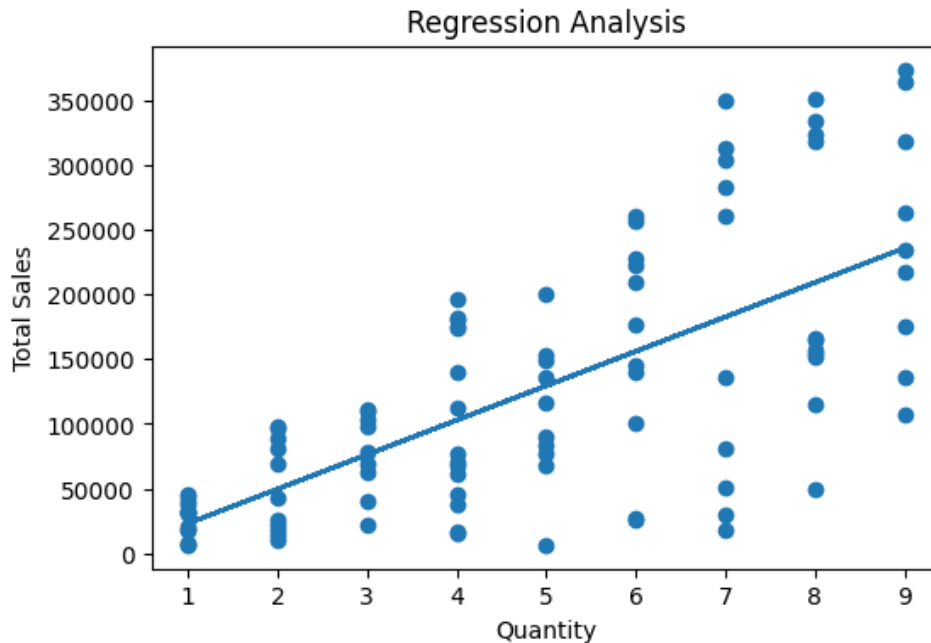
Margin of Error: 19874.13

## 8. Regression Analysis

Regression Intercept: -3638.74

Regression Coefficient: 26629.54

R-squared Value: 0.47



## 9. Business Insights and Recommendations

Higher quantity sold leads to higher total sales.

Product categories significantly influence sales performance.

Sales distribution shows variation across regions.

Regression analysis indicates quantity is a strong predictor of sales.

Correlation analysis suggests strong relationships among sales variables.

## 10. Conclusion

This project demonstrates practical business analytics using statistical techniques. The analysis provides actionable insights that can help improve sales strategy, product planning, and business decision-making.